Signal, Image and Video Processing (2021) 15:103–110
https://doi.org/10.1007/s11760-020-01728-x

ORIGINAL PAPER

A new kind of convolution, correlation and product theorems related to quaternion linear canonical transform

Zhen-Wei Li[1,2] · Wen-Biao Gao[1,2] · Bing-Zhao Li[1,2]

Received: 4 March 2020 / Revised: 11 May 2020 / Accepted: 13 June 2020 / Published online: 3 July 2020

Abstract
The quaternion linear canonical transform (QLCT) has been investigated and used for many years by various research communities; it has been shown to be a powerful tool for signal processing. In this paper, a novel canonical convolution operator and a related correlation operator for QLCT are proposed. Moreover, based on the proposed operators, the corresponding generalized convolution, correlation and product theorems are studied.

Keywords Linear canonical transform · Quaternion linear canonical transform · Convolution operator · Correlation operator · Product theorem

1 Introduction

The linear canonical transform (LCT) is a linear integral transformation and has been proved to be a powerful tool in several areas, including optics, signal processing, optimal filtering and time–frequency analysis [1–6]. The LCT has been found as a generalization of many mathematical transforms, for instance, Fourier, Laplace, fractional Fourier and other transformations. Many useful properties of the LCT have been established, such as convolution, correlation, product, sampling and uncertainty principles [7–13].

Recently, extending the integral transforms from real and complex domain to quaternion domain has become more and more popular [14–19]. As a generalization of the LCT, the quaternion linear canonical transform (QLCT) has received much attention from researchers and obtained many interesting results. In [16,17], Kou et al. first introduced the QLCT which is obtained by replacing the Fourier kernel with the quaternion Fourier transform kernel in the LCT definition. Then, many important properties, such as the Parseval's formula, uncertainty principles, convolution theorems and reconstruction formula associated with the QLCT are discussed [16,17,20–23].

Among these properties, convolution, correlation and product theorems are fundamental in the theory of linear time-invariant system and are important mathematical operations with several applications in signal processing, optics and filter design [10–13,24–28]. On the one hand, the QLCT has become one of the research hotspots in the field signal processing. On the other hand, the convolution, correlation and product are important properties of QLCT. However, there are few studies for establishing the convolution, correlation and product theorems associated with the QLCT as far as we know. Therefore, it is worthwhile and interesting to study the convolution and related theorems associated with the QLCT, which can be useful in signal processing theory and application.

The main objective of this paper is to propose a novel canonical convolution operator and a related correlation operator for QLCT and study the corresponding generalized convolution, correlation and product theorems. The paper is organized as follows: In Sect. 2, we provides a brief introduction of quaternion algebra, the definition of QLCT and the relationship between QLCT and quaternion Fourier transform (QFT). In Sect. 3, we first propose a novel canonical convolution operator for the QLCT and study the corresponding generalized convolution theorem. Then, we propose a

✉ Bing-Zhao Li
li_bingzhao@bit.edu.cn

Zhen-Wei Li
lizhenweibit@163.com

Wen-Biao Gao
wenbiaogao@163.com

[1] School of Mathematics and Statistics, Beijing Institute of Technology, Beijing 100081, China

[2] Beijing Key Laboratory on MCAACI, Beijing Institute of Technology, Beijing 100081, China

correlation operator for QLCT and investigate the corresponding correlation theorem. Finally, based on the definition of the QLCT, we research the product theorem associated with QLCT. In Sect. 4, we conclude the paper.

2 Preliminaries

2.1 Quaternion algebra

The quaternion, which is a type of hypercomplex number, was formally introduced by Hamilton in 1843 [29] and is denoted by $\mathbb{H}$. Every element of $\mathbb{H}$ can be written in the following form

$$\mathbb{H} = \{q = q_0 + q_1\mathbf{i} + q_2\mathbf{j} + q_3\mathbf{k};\ q_0, q_1, q_2, q_3 \in \mathbb{R}\}, \quad (1)$$

where $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are three different imaginary parts and obey the following multiplication rules

$$\begin{aligned} &\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1, \\ &\mathbf{ij} = -\mathbf{ji} = \mathbf{k}, \quad \mathbf{jk} = -\mathbf{kj} = \mathbf{i}, \quad \mathbf{ki} = -\mathbf{ik} = \mathbf{j}. \end{aligned} \quad (2)$$

For a quaternion $q = q_0 + q_1\mathbf{i} + q_2\mathbf{j} + q_3\mathbf{k} \in \mathbb{H}$, q_0 is the scalar part of q and denoted by $S(q)$, $q_1\mathbf{i} + q_2\mathbf{j} + q_3\mathbf{k}$ is the pure part of q and denoted by $V(q)$. If $q_0 = 0$, then q is called pure quaternion.

The conjugate of a quaternion q is: $\bar{q} = q_0 - q_1\mathbf{i} - q_2\mathbf{j} - q_3\mathbf{k}$. For any two quaternions p and q, the multiplication of their conjugate satisfies: $\overline{p \cdot q} = \bar{q} \cdot \bar{p}$.

The norm of quaternion q is: $|q| = \sqrt{q \cdot \bar{q}}$. If q has a unit norm ($|q| = 1$), then q is called unit quaternion. The inverse of $q \in \mathbb{H} \setminus \{0\}$ is: $q^{-1} = \dfrac{\bar{q}}{|q|^2}$.

A quaternion-valued function $f : \mathbb{R}^2 \to \mathbb{H}$ can be expressed as

$$f(\mathbf{x}) = f_0(\mathbf{x}) + f_1(\mathbf{x})\mathbf{i} + f_2(\mathbf{x})\mathbf{j} + f_3(\mathbf{x})\mathbf{k}, \quad (3)$$

where $f_0(\mathbf{x}),\ f_1(\mathbf{x}),\ f_2(\mathbf{x}),\ f_3(\mathbf{x}) \in \mathbb{R}$.

2.2 The quaternion linear canonical transform

A special linear group is denoted as

$$\mathrm{SL}(2, R) = A \in M(2, \mathbb{R}|\det(A) = 1), \quad (4)$$

where $M(2, \mathbb{R})$ is a set of 2×2 real matrices.

The two-sided QLCT of $f \in L^1(\mathbb{R}^2, \mathbb{H})$ is defined by [16,17]

$$\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f\}(\omega) = \int_{\mathbb{R}^2} K^{\mathbf{i}}_{A_1}(x_1, \omega_1) f(\mathbf{x}) K^{\mathbf{j}}_{A_2}(x_2, \omega_2)\mathrm{d}\mathbf{x}, \quad (5)$$

where $A_1,\ A_2 \in \mathrm{SL}(2, R)$ and the kernel functions $K^{\mathbf{i}}_{A_1}(x_1, \omega_1)$ and $K^{\mathbf{j}}_{A_2}(x_2, \omega_2)$ of the QLCT are given by

$$K^{\mathbf{i}}_{A_1}(x_1, \omega_1) = \begin{cases} \dfrac{1}{\sqrt{2\pi b_1 \mathbf{i}}} e^{\mathbf{i}\left(\frac{a_1}{2b_1}x_1^2 - \frac{1}{b_1}x_1\omega_1 + \frac{d_1}{2b_1}\omega_1^2\right)}, & b_1 \neq 0 \\ \sqrt{d_1} e^{\mathbf{i}\frac{c_1 d_1}{2}\omega_1^2}, & b_1 = 0 \end{cases}, \quad (6)$$

and

$$K^{\mathbf{j}}_{A_2}(x_2, \omega_2) = \begin{cases} \dfrac{1}{\sqrt{2\pi b_2 \mathbf{j}}} e^{\mathbf{j}\left(\frac{a_2}{2b_2}x_2^2 - \frac{1}{b_2}x_2\omega_2 + \frac{d_2}{2b_2}\omega_2^2\right)}, & b_2 \neq 0 \\ \sqrt{d_2} e^{\mathbf{j}\frac{c_2 d_2}{2}\omega_2^2}, & b_2 = 0 \end{cases}. \quad (7)$$

Here, $\mathbf{x} = (x_1, x_2)$, $\omega = (\omega_1, \omega_2) \in \mathbb{R}^2$. From the above definition, the QLCT of a signal is essentially a chirp multiplication when $b_i = 0, i = 1, 2$ and it has no particular interest for our objective in this paper. Hence, we will study for the case $b_1 b_2 \neq 0$ in the following sections unless otherwise specified.

Moreover, the QLCT above is invertible under some suitable conditions, and the inversion transform of the QLCT is as follows:

Suppose $f \in L^1(\mathbb{R}^2, \mathbb{H})$, then the inversion of the two-sided QLCT of f is given by [16,17]

$$\begin{aligned} f(\mathbf{x}) &= \left\{\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\right\}^{-1}\{f\}(\mathbf{x}) \\ &= \int_{\mathbb{R}^2} K^{-\mathbf{i}}_{A_1}(x_1, \omega_1)\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f\}(\omega) K^{-\mathbf{j}}_{A_2}(x_2, \omega_2)\mathrm{d}\omega. \end{aligned} \quad (8)$$

2.3 Relationship between the QLCT and QFT

Recently, the fundamental relationship between the QLCT and QFT is derived [30]. Without explanation, $\mathscr{F}$ and $\mathscr{L}^{\mathbb{H}}_{A_1,A_2}$ denote the two-sided QFT and two-sided QLCT operator, respectively. The QLCT can be reduced to QFT as follows:

$$\mathscr{F}\{g\}(\omega) = \frac{1}{\sqrt{(2\pi)^2}} \int_{\mathbb{R}^2} e^{-\mathbf{i}x_1\omega_1} g(\mathbf{x}) e^{-\mathbf{j}x_2\omega_2}\mathrm{d}\mathbf{x}, \quad (9)$$

where

$$\begin{aligned} &\mathscr{F}\{g\}(\omega) = \widetilde{\mathscr{F}}(\mathbf{b}\omega), \\ &g(\mathbf{x}) = \frac{1}{\sqrt{b_1\mathbf{i}}}\widetilde{f}(\mathbf{x})\frac{1}{\sqrt{b_2\mathbf{j}}}, \quad \widetilde{f}(\mathbf{x}) = e^{\mathbf{i}\frac{a_1}{2b_1}x_1^2} f(\mathbf{x}) e^{\mathbf{j}\frac{a_2}{2b_2}x_2^2}, \end{aligned} \quad (10)$$

with

$$\widetilde{\mathscr{F}}(\omega)=\frac{1}{\sqrt{(2\pi)^2}}\int_{\mathbb{R}^2}e^{-\mathbf{i}\frac{\omega_1}{b_1}x_1}g(\mathbf{x})e^{-\mathbf{j}\frac{\omega_2}{b_2}x_2}\mathrm{d}\mathbf{x}$$
$$=e^{-\mathbf{i}\frac{d_1}{2b_1}\omega_1^2}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f\}(\omega)e^{-\mathbf{j}\frac{d_2}{2b_2}\omega_2^2}. \tag{11}$$

Herein, $\mathbf{b}=(b_1,b_2)\in\mathbb{R}^2$, the elementwise multiplication $\mathbf{b}\omega=(b_1\omega_1,b_2\omega_2)$, and the elementwise division $\frac{\omega}{\mathbf{b}}=\left(\frac{\omega_1}{b_1},\frac{\omega_2}{b_2}\right)$. The specific proof of this relationship is in [5].

3 Main results

In this section, we first introduce a new convolution operator which is an extension of the convolution definition from the one-dimensional LCT (see [11]) to the two-dimensional QLCT domain, and the corresponding generalized convolution theorem is researched. It is shown that our new canonical convolution operator is much more flexible and useful in certain cases. Then, a correlation operator for QLCT which is related to the convolution operator is proposed, and the corresponding correlation theorem is investigated. Finally, based on the definition of the QLCT, we research the product theorem associated with QLCT.

3.1 Convolution theorem associated with QLCT

Definition 1 For any two functions $f,g\in L^1(\mathbb{R}^2;\mathbb{H})$, the convolution operator $\circledast$ for the QLCT is defined as

$$(f\circledast g)(\mathbf{x})$$
$$=\int_{\mathbb{R}^2}\frac{1}{\sqrt{2\pi b_1\mathbf{i}}}e^{\mathbf{i}\left(\frac{a_1}{b_1}u_1^2-\frac{a_1}{b_1}u_1x_1+a_1x_1-a_1u_1\right)}f(\mathbf{u})$$
$$\cdot g(\mathbf{x}-\mathbf{u}+\mathbf{b})\frac{1}{\sqrt{2\pi b_2\mathbf{j}}}e^{\mathbf{j}\left(\frac{a_2}{b_2}u_2^2-\frac{a_2}{b_2}u_2x_2+a_2x_2-a_2u_2\right)}\mathrm{d}\mathbf{u}. \tag{12}$$

Based on the classical convolution operator $*$, the new convolution in (12) can be expressed as two different forms as following

$$h(\mathbf{x}):=(f\circledast g)(\mathbf{x})$$
$$=\left(e^{\mathbf{i}\frac{a_1}{2b_1}t_1^2}\cdot f(\mathbf{t})\cdot e^{\mathbf{j}\frac{a_2}{2b_2}t_2^2}\right)$$
$$*\left(e^{\mathbf{i}(a_1t_1+\frac{a_1}{2b_1}t_1^2)}\cdot g(\mathbf{t}+\mathbf{b})\cdot e^{\mathbf{j}(a_2t_2+\frac{a_2}{2b_2}t_2^2)}\right)(\mathbf{x})$$
$$\times\frac{1}{\sqrt{4\pi^2b_1b_2\mathbf{ij}}}e^{-\mathbf{i}\frac{a_1}{2b_1}x_1^2}e^{-\mathbf{j}\frac{a_2}{2b_2}x_2^2}, \tag{13}$$

$$h(\mathbf{x}):=(f\circledast g)(\mathbf{x})$$
$$=\left(e^{\mathbf{i}(-a_1t_1+\frac{a_1}{2b_1}t_1^2)}\cdot f(\mathbf{t})\cdot e^{\mathbf{j}(-a_2t_2+\frac{a_2}{2b_2}t_2^2)}\right)$$
$$*\left(e^{\mathbf{i}\frac{a_1}{2b_1}t_1^2}\cdot g(\mathbf{t}+\mathbf{b})\cdot e^{\mathbf{j}\frac{a_2}{2b_2}t_2^2}\right)(\mathbf{x})$$
$$\times\frac{1}{\sqrt{4\pi^2b_1b_2\mathbf{ij}}}e^{\mathbf{i}(a_1x_1-\frac{a_1}{2b_1}x_1^2)}e^{\mathbf{j}(a_2x_2-\frac{a_2}{2b_2}x_2^2)}. \tag{14}$$

Moreover, these two different implementations for the new convolution operator $\circledast$ are shown in Figs. 1 and 2, respectively.

On the one hand, it is easy to see that the new convolution operator $\circledast$ is a single integral which is much simpler than the triple integral mentioned in [24]. On the other hand, the new convolution operator $\circledast$ has two different implementations in filter design due to Eqs. (13) and (14). Therefore, in certain cases, the new convolution operator proposed in this paper is much more flexible and useful than those in [2,22–25,27].

After simple computation, for any $f,g,h\in L^1(\mathbb{R}^2;\mathbb{H})$, the convolution operator $\circledast$ satisfies associative property and distributive property are as following

(1) Associativity: $(f\circledast g)\circledast h=f\circledast(g\circledast h)$,

(2) Distributivity: $f\circledast(g+h)=f\circledast g+f\circledast h$.

Based on the definition and properties of the new convolution operator $\circledast$, the convolution theorem associated with the QLCT is derived as follows:

Theorem 1 *Let $f,g\in L^1(\mathbb{R}^2;\mathbb{H})$ be two quaternion-valued signal, then the convolution theorem associated with the QLCT is given by*

$$\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f\circledast g\}(\omega)$$
$$=\Phi(\boldsymbol{i})[\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f_0\}(\omega)+\boldsymbol{i}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f_1\}(\omega)$$
$$+\boldsymbol{j}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f_2\}(\omega)+\boldsymbol{k}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f_3\}(\omega)]$$
$$\cdot[\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{g_0\}(\omega)+\boldsymbol{i}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{g_1\}(\omega)$$
$$+\boldsymbol{j}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{g_2\}(\omega)+\boldsymbol{k}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{g_3\}(\omega)]\cdot\Phi(\boldsymbol{j}), \tag{15}$$

where $\Phi(\mathbf{i})=e^{\mathbf{i}\left(\omega_1-\frac{d_1}{2b_1}\omega_1^2-\frac{a_1b_1}{2}\right)}$, $\Phi(\mathbf{j})=e^{\mathbf{j}\left(\omega_2-\frac{d_2}{2b_2}\omega_2^2-\frac{a_2b_2}{2}\right)}$.

Proof From Eqs. (5) and (12), we have

$$\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f\circledast g\}(\omega)$$
$$=\int_{\mathbb{R}^2}\int_{\mathbb{R}^2}\frac{1}{2\pi b_1\mathbf{i}}e^{\mathbf{i}\left(\frac{a_1}{2b_1}x_1^2-\frac{1}{b_1}x_1\omega_1+\frac{d_1}{2b_1}\omega_1^2\right)}$$
$$\cdot e^{\mathbf{i}\left(\frac{a_1}{b_1}u_1^2-\frac{a_1}{b_1}u_1x_1+a_1x_1-a_1u_1\right)}f(\mathbf{u})g(\mathbf{x}-\mathbf{u}+\mathbf{b})\frac{1}{2\pi b_2\mathbf{j}}$$
$$\cdot e^{\mathbf{j}\left(\frac{a_2}{2b_2}x_2^2-\frac{1}{b_2}x_2\omega_2+\frac{d_2}{2b_2}\omega_2^2\right)}e^{\mathbf{j}\left(\frac{a_2}{b_2}u_2^2-\frac{a_2}{b_2}u_2x_2+a_2x_2-a_2u_2\right)}\mathrm{d}\mathbf{x}\mathrm{d}\mathbf{u}$$

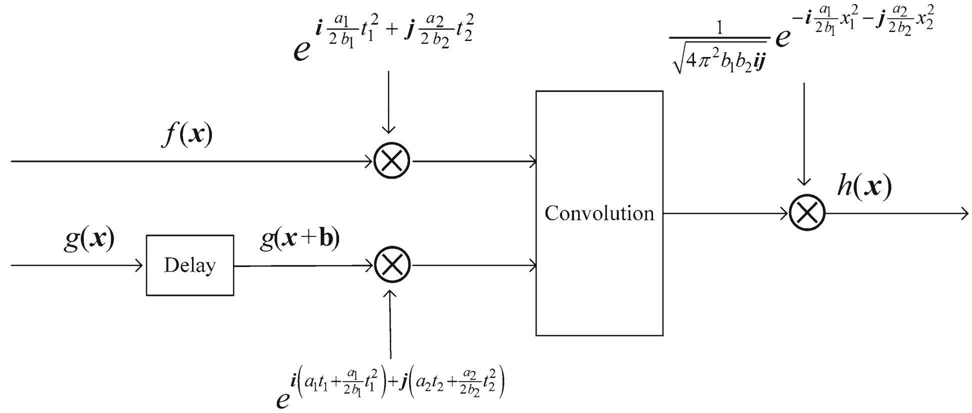

Fig. 1 One expression of the convolution operator $\circledast$

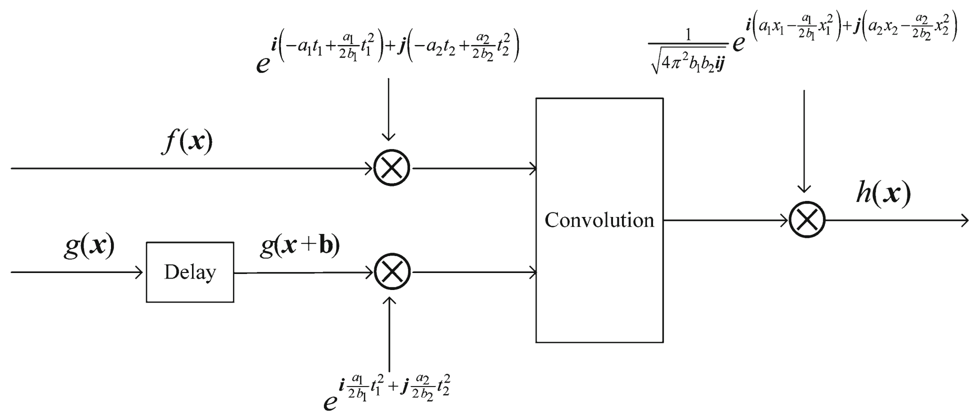

Fig. 2 Another expression of the convolution operator $\circledast$

$$
\begin{aligned}
&= \int_{\mathbb{R}^2}\int_{\mathbb{R}^2} \frac{1}{2\pi b_1 \mathbf{i}} \cdot e^{-\mathbf{i}\frac{a_1 b_1}{2}} \\
&\cdot e^{\mathbf{i}\frac{a_1}{2b_1}\left(2u_1^2-2x_1u_1+x_1^2+2b_1x_1-2b_1u_1+b_1^2\right)-\mathbf{i}\frac{1}{b_1}x_1\omega_1+\mathbf{i}\frac{d_1}{2b_1}\omega_1^2} \\
&\cdot f(\mathbf{u})g(\mathbf{x}-\mathbf{u}+\mathbf{b})\frac{1}{2\pi b_2 \mathbf{j}}e^{-\mathbf{j}\frac{a_2 b_2}{2}} \\
&\cdot e^{\mathbf{j}\frac{a_2}{2b_2}\left(2u_2^2-2x_2u_2+x_2^2+2b_2x_1-2b_2u_2+b_2^2\right)-\mathbf{j}\frac{1}{b_2}x_2\omega_2+\mathbf{j}\frac{d_2}{2b_2}\omega_2^2}\mathrm{d}\mathbf{x}\mathrm{d}\mathbf{u} \\
&= \frac{1}{2\pi b_1 \mathbf{i}}e^{-\mathbf{i}\frac{a_1 b_1}{2}} \\
&\cdot \int_{\mathbb{R}^2}\int_{\mathbb{R}^2} e^{\mathbf{i}\frac{a_1}{2b_1}\left[u_1^2+(x_1-u_1+b_1)^2\right]-\mathbf{i}\frac{1}{b_1}x_1\omega_1+\mathbf{i}\frac{d_1}{2b_1}\omega_1^2} f(\mathbf{u}) \\
&\cdot g(\mathbf{x}-\mathbf{u}+\mathbf{b})e^{\mathbf{j}\frac{a_2}{2b_2}\left[u_2^2+(x_2-u_2+b_2)^2\right]-\mathbf{j}\frac{1}{b_2}x_2\omega_2+\mathbf{j}\frac{d_2}{2b_2}\omega_2^2}\mathrm{d}\mathbf{x}\mathrm{d}\mathbf{u} \\
&\cdot \frac{1}{2\pi b_2 \mathbf{j}}e^{-\mathbf{j}\frac{a_2 b_2}{2}}.
\end{aligned}
\tag{16}
$$

Let $\mathbf{x}-\mathbf{u}+\mathbf{b}=\mathbf{y}$, Eq. (16) can be rewritten as

$$
\begin{aligned}
&\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f \circledast g\}(\omega) \\
&= \frac{1}{2\pi b_1 \mathbf{i}}e^{-\mathbf{i}\frac{a_1 b_1}{2}} \\
&\cdot \int_{\mathbb{R}^2}\int_{\mathbb{R}^2} e^{\mathbf{i}\frac{a_1}{2b_1}\left(u_1^2+y_1^2\right)-\mathbf{i}\frac{1}{b_1}\omega_1(u_1+y_1-b_1)+\mathbf{i}\frac{d_1}{2b_1}\omega_1^2} \\
&\cdot f(\mathbf{u})g(\mathbf{y})e^{\mathbf{j}\frac{a_2}{2b_2}\left(u_2^2+y_2^2\right)-\mathbf{j}\frac{1}{b_2}\omega_2(u_2+y_2-b_2)+\mathbf{j}\frac{d_2}{2b_2}\omega_2^2}\mathrm{d}\mathbf{y}\mathrm{d}\mathbf{u} \\
&\cdot \frac{1}{2\pi b_2 \mathbf{j}}e^{-\mathbf{j}\frac{a_2 b_2}{2}} \\
&= \frac{1}{2\pi b_1 \mathbf{i}}e^{-\mathbf{i}\frac{a_1 b_1}{2}} \\
&\cdot \int_{\mathbb{R}^2}\int_{\mathbb{R}^2} e^{\mathbf{i}\frac{a_1}{2b_1}\left(u_1^2+y_1^2\right)-\mathbf{i}\frac{1}{b_1}\omega_1(u_1+y_1-b_1)+\mathbf{i}\frac{d_1}{2b_1}\omega_1^2} \\
&\cdot \left[f_0(\mathbf{u})+\mathbf{i}f_1(\mathbf{u})+\mathbf{j}f_2(\mathbf{u})+\mathbf{k}f_3(\mathbf{u})\right] \\
&\cdot \left[g_0(\mathbf{y})+\mathbf{i}g_1(\mathbf{y})+\mathbf{j}g_2(\mathbf{y})+\mathbf{k}g_3(\mathbf{y})\right] \\
&\cdot e^{\mathbf{j}\frac{a_2}{2b_2}\left(u_2^2+y_2^2\right)-\mathbf{j}\frac{1}{b_2}\omega_2(u_2+y_2-b_2)+\mathbf{j}\frac{d_2}{2b_2}\omega_2^2}\mathrm{d}\mathbf{y}\mathrm{d}\mathbf{u} \\
&\cdot \frac{1}{2\pi b_2 \mathbf{j}}e^{-\mathbf{j}\frac{a_2 b_2}{2}} \\
&= \frac{1}{2\pi b_1 \mathbf{i}}e^{\mathbf{i}\left(\omega_1-\frac{d_1}{2b_1}\omega_1^2-\frac{a_1 b_1}{2}\right)} \\
&\cdot \int_{\mathbb{R}^2}\int_{\mathbb{R}^2} e^{\mathbf{i}\frac{a_1}{2b_1}\left(u_1^2+y_1^2\right)-\mathbf{i}\frac{1}{b_1}\omega_1(u_1+y_1)+\mathbf{i}\frac{d_1}{b_1}\omega_1^2} \\
&\cdot \left[f_0(\mathbf{u})+\mathbf{i}f_1(\mathbf{u})+\mathbf{j}f_2(\mathbf{u})+\mathbf{k}f_3(\mathbf{u})\right] \\
&\cdot \left[g_0(\mathbf{y})+\mathbf{i}g_1(\mathbf{y})+\mathbf{j}g_2(\mathbf{y})+\mathbf{k}g_3(\mathbf{y})\right] \\
&\cdot e^{\mathbf{j}\frac{a_2}{2b_2}\left(u_2^2+y_2^2\right)-\mathbf{j}\frac{1}{b_2}\omega_2(u_2+y_2)+\mathbf{j}\frac{d_2}{b_2}\omega_2^2}\mathrm{d}\mathbf{y}\mathrm{d}\mathbf{u} \\
&\cdot \frac{1}{2\pi b_2 \mathbf{j}}e^{\mathbf{j}\left(\omega_2-\frac{d_2}{2b_2}\omega_2^2-\frac{a_2 b_2}{2}\right)}.
\end{aligned}
\tag{17}
$$

For convenience, we denote that

$$\Phi(\mathbf{i}) = e^{\mathbf{i}\left(\omega_1 - \frac{d_1}{2b_1}\omega_1^2 - \frac{a_1 b_1}{2}\right)},$$
$$\Phi(\mathbf{j}) = e^{\mathbf{j}\left(\omega_2 - \frac{d_2}{2b_2}\omega_2^2 - \frac{a_2 b_2}{2}\right)}. \tag{18}$$

Thus, Eq. (17) can be rewritten as

$$\begin{aligned}
&\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f \circledast g\}(\omega) \\
&= \Phi(\mathbf{i}) \int_{\mathbb{R}^2} \frac{1}{\sqrt{2\pi b_1 \mathbf{i}}} e^{\mathbf{i}\left(\frac{a_1}{2b_1}u_1^2 - \frac{1}{b_1}\omega_1 u_1 + \frac{d_1}{2b_1}\omega_1^2\right)} \\
&\quad \cdot [f_0(\mathbf{u}) + \mathbf{i} f_1(\mathbf{u}) + \mathbf{j} f_2(\mathbf{u}) + \mathbf{k} f_3(\mathbf{u})] \\
&\quad \cdot \frac{1}{\sqrt{2\pi b_2 \mathbf{j}}} e^{\mathbf{j}\left(\frac{a_2}{2b_2}u_2^2 - \frac{1}{b_2}\omega_2 u_2 + \frac{d_2}{2b_2}\omega_2^2\right)} \mathbf{du} \\
&\quad \cdot \int_{\mathbb{R}^2} \frac{1}{\sqrt{2\pi b_1 \mathbf{i}}} e^{\mathbf{i}\left(\frac{a_1}{2b_1}y_1^2 - \frac{1}{b_1}\omega_1 y_1 + \frac{d_1}{2b_1}\omega_1^2\right)} \\
&\quad \cdot \left[g_0(\mathbf{y}) + \mathbf{i} g_1(\mathbf{y}) + \mathbf{j} g_2(\mathbf{y}) + \mathbf{k} g_3(\mathbf{y})\right] \\
&\quad \cdot \frac{1}{\sqrt{2\pi b_2 \mathbf{j}}} e^{\mathbf{j}\left(\frac{a_2}{2b_2}y_2^2 - \frac{1}{b_2}\omega_2 y_2 + \frac{d_2}{2b_2}\omega_2^2\right)} \mathbf{dy} \cdot \Phi(\mathbf{j}) \\
&= \Phi(\mathbf{i})[\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f_0\}(\omega) + \mathbf{i}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f_1\}(\omega) \\
&\quad + \mathbf{j}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f_2\}(\omega) + \mathbf{k}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f_3\}(\omega)] \\
&\quad \cdot [\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{g_0\}(\omega) + \mathbf{i}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{g_1\}(\omega) \\
&\quad + \mathbf{j}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{g_2\}(\omega) + \mathbf{k}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{g_3\}(\omega)] \cdot \Phi(\mathbf{j}).
\end{aligned} \tag{19}$$

This completes the proof. □

Remark 1 If we use the matrix parameters $A_1 = A_2 = [0,\ 1;\ -1,\ 0]$, then Eq. (12) will lead to

$$\begin{aligned}
&\mathscr{F}\{f \circledast g\}(\omega) \\
&= e^{\mathbf{i}\omega_1}\{[\mathscr{F}\{f_0\}(\omega)\mathscr{F}\{g_0\}(\omega) - \mathscr{F}\{f_1\}(\omega)\mathscr{F}\{g_1\}(\omega) \\
&\quad - \mathscr{F}\{f_2\}(\omega)\mathscr{F}\{g_2\}(\omega) - \mathscr{F}\{f_3\}(\omega)\mathscr{F}\{g_3\}(\omega)] \\
&\quad + \mathbf{i}[\mathscr{F}\{f_0\}(\omega)\mathscr{F}\{g_1\}(\omega) + \mathscr{F}\{f_1\}(\omega)\mathscr{F}\{g_0\}(\omega) \\
&\quad + \mathscr{F}\{f_2\}(\omega)\mathscr{F}\{g_3\}(\omega) - \mathscr{F}\{f_3\}(\omega)\mathscr{F}\{g_2\}(\omega)] \\
&\quad + \mathbf{j}[\mathscr{F}\{f_0\}(\omega)\mathscr{F}\{g_2\}(\omega) + \mathscr{F}\{f_2\}(\omega)\mathscr{F}\{g_0\}(\omega) \\
&\quad + \mathscr{F}\{f_3\}(\omega)\mathscr{F}\{g_1\}(\omega) - \mathscr{F}\{f_1\}(\omega)\mathscr{F}\{g_3\}(\omega)] \\
&\quad + \mathbf{k}[\mathscr{F}\{f_0\}(\omega)\mathscr{F}\{g_3\}(\omega) + \mathscr{F}\{f_3\}(\omega)\mathscr{F}\{g_0\}(\omega) \\
&\quad + \mathscr{F}\{f_1\}(\omega)\mathscr{F}\{g_2\}(\omega) - \mathscr{F}\{f_2\}(\omega)\mathscr{F}\{g_1\}(\omega)]\} \cdot e^{\mathbf{j}\omega_2}.
\end{aligned} \tag{20}$$

Observe that the above form is quite similar to the convolution theorem associated with the QFT [30].

In addition, it is worth noting that although the convolution and correlation formula are similar, their physical meaning is quite different. Convolution is used to describe the relationship between the inputs and outputs of a linear time-invariant system. However, the correlation operation is used to analyze the correlation between two signals. Because of the different physical meaning, the applications of convolution and correlation are also different. Convolution is generally applied to the filtering, sampling and image encryption. Correlation is generally applied to the signal detection, pattern recognition. Therefore, the study of correlation is also very meaningful.

3.2 Correlation theorem for QLCT

In this subsection, a correlation operator for QLCT is proposed, and the corresponding correlation theorem is investigated.

Definition 2 For any two functions $f, g \in L^1(\mathbb{R}^2; \mathbb{H})$, the correlation operator $\triangle$ of the QLCT is defined as follows

$$\begin{aligned}
&(f \triangle g)(\mathbf{x}) \\
&= \int_{\mathbb{R}^2} \frac{1}{\sqrt{2\pi b_1 \mathbf{i}}} e^{\mathbf{i}\left(\frac{a_1}{b_1}u_1^2 + \frac{a_1}{b_1}u_1 x_1 + a_1 x_1 + a_1 u_1\right)} \overline{f(\mathbf{u})} \\
&\quad \cdot g(\mathbf{x} + \mathbf{u} + \mathbf{b}) \frac{1}{\sqrt{2\pi b_2 \mathbf{j}}} e^{\mathbf{j}\left(\frac{a_2}{b_2}u_2^2 + \frac{a_2}{b_2}u_2 x_2 + a_2 x_2 + a_2 u_2\right)} \mathbf{du}.
\end{aligned} \tag{21}$$

Similarly to the convolution operator $\circledast$ proposed in Sect. 3.1, the correlation operator $\triangle$ also satisfies associative and distributive property.

(1) Associativity: $(f \triangle g) \triangle h = f \triangle (g \triangle h)$,
(2) Distributivity: $f \triangle (g + h) = f \triangle g + f \triangle h$.

Based on the definition and properties of the new correlation operator $\triangle$, the correlation theorem associated with the QLCT is derived as follows:

Theorem 2 *Let $f, g \in L^1(\mathbb{R}^2; \mathbb{H})$, the QLCT of the convolution of f and g is given by*

$$\begin{aligned}
&\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f \triangle g\}(\omega) \\
&= \Phi(\boldsymbol{i})[\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f_0\}(-\omega) - \boldsymbol{i}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f_1\}(-\omega) \\
&\quad - \boldsymbol{j}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f_2\}(-\omega) - \boldsymbol{k}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f_3\}(-\omega)] \\
&\quad \cdot [\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{g_0\}(\omega) + \boldsymbol{i}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{g_1\}(\omega) + \boldsymbol{j}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{g_2\}(\omega) \\
&\quad + \boldsymbol{k}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{g_3\}(\omega)] \cdot \Phi(\boldsymbol{j}).
\end{aligned} \tag{22}$$

Proof Let $\mathbf{u} = -\mathbf{v}$, Eq. (22) can be rewritten as

$$\begin{aligned}
&(f \triangle g)(\mathbf{x}) \\
&= -\int_{\mathbb{R}^2} \frac{1}{\sqrt{2\pi b_1 \mathbf{i}}} e^{\mathbf{i}\left(\frac{a_1}{b_1}v_1^2 - \frac{a_1}{b_1}v_1 x_1 + a_1 x_1 - a_1 v_1\right)} \overline{f(-\mathbf{v})} \\
&\quad \cdot g(\mathbf{x} - \mathbf{v} + \mathbf{b}) \frac{1}{\sqrt{2\pi b_2 \mathbf{j}}} e^{\mathbf{j}\left(\frac{a_2}{b_2}v_2^2 - \frac{a_2}{b_2}v_2 x_2 + a_2 x_2 - a_2 v_2\right)} \mathbf{dv}.
\end{aligned} \tag{23}$$

Let $h(\mathbf{v}) = -\overline{f(-\mathbf{v})}$, we have

$$\begin{aligned}
&\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{h\}(\omega)\\
&= \int_{\mathbb{R}^2} \frac{1}{\sqrt{2\pi b_1 \mathbf{i}}} e^{\mathbf{i}\left(\frac{a_1}{2b_1}v_1^2 - \frac{1}{b_1}\omega_1 v_1 + \frac{d_1}{2b_1}\omega_1^2\right)} h(\mathbf{v})\\
&\cdot \frac{1}{\sqrt{2\pi b_2 \mathbf{j}}} e^{\mathbf{j}\left(\frac{a_2}{2b_2}v_2^2 - \frac{1}{b_2}\omega_2 v_2 + \frac{d_2}{2b_2}\omega_2^2\right)} d\mathbf{v}\\
&= -\int_{\mathbb{R}^2} \frac{1}{\sqrt{2\pi b_1 \mathbf{i}}} e^{\mathbf{i}\left(\frac{a_1}{2b_1}v_1^2 - \frac{1}{b_1}\omega_1 v_1 + \frac{d_1}{2b_1}\omega_1^2\right)} \overline{f(-\mathbf{v})}\\
&\cdot \frac{1}{\sqrt{2\pi b_2 \mathbf{j}}} e^{\mathbf{j}\left(\frac{a_2}{2b_2}v_2^2 - \frac{1}{b_2}\omega_2 v_2 + \frac{d_2}{2b_2}\omega_2^2\right)} d\mathbf{v}
\end{aligned}$$

$$\begin{aligned}
&= \int_{\mathbb{R}^2} \frac{1}{\sqrt{2\pi b_1 \mathbf{i}}} e^{\mathbf{i}\left(\frac{a_1}{2b_1}u_1^2 - \frac{1}{b_1}(-\omega_1) u_1 + \frac{d_1}{2b_1}\omega_1^2\right)} \overline{f(\mathbf{u})}\\
&\cdot \frac{1}{\sqrt{2\pi b_2 \mathbf{j}}} e^{\mathbf{j}\left(\frac{a_2}{2b_2}u_2^2 - \frac{1}{b_2}(-\omega_2) u_2 + \frac{d_2}{2b_2}\omega_2^2\right)} d\mathbf{u}\\
&= \mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{\bar{f}\}(-\omega)\\
&= \mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f_0\}(-\omega) - \mathbf{i}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f_1\}(-\omega)\\
&- \mathbf{j}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f_2\}(-\omega) - \mathbf{k}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f_0\}(-\omega)
\end{aligned} \tag{24}$$

and Eq. (23) can be rewritten as

$$\begin{aligned}
&(f \vartriangle g)(\mathbf{x})\\
&= \int_{\mathbb{R}^2} \frac{1}{\sqrt{2\pi b_1 \mathbf{i}}} e^{\mathbf{i}\left(\frac{a_1}{b_1}v_1^2 - \frac{a_1}{b_1}v_1 x_1 + a_1 x_1 - a_1 v_1\right)} h(\mathbf{v})\\
&\cdot g(\mathbf{x} - \mathbf{v} + \mathbf{b}) \frac{1}{\sqrt{2\pi b_2 \mathbf{j}}} e^{\mathbf{j}\left(\frac{a_2}{b_2}v_2^2 - \frac{a_2}{b_2}v_2 x_2 + a_2 x_2 - a_2 v_2\right)} d\mathbf{v}.
\end{aligned} \tag{25}$$

By Eqs. (12) (19) and (24), we have

$$\begin{aligned}
&\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f \vartriangle g\}(\omega)\\
&= \Phi(\mathbf{i})[\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f_0\}(-\omega) - \mathbf{i}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f_1\}(-\omega)\\
&- \mathbf{j}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f_2\}(-\omega) - \mathbf{k}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{f_3\}(-\omega)]\\
&\cdot [\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{g_0\}(\omega) + \mathbf{i}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{g_1\}(\omega) + \mathbf{j}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{g_2\}(\omega)\\
&+ \mathbf{k}\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{g_3\}(\omega)] \cdot \Phi(\mathbf{j}),
\end{aligned} \tag{26}$$

where $\Phi(\mathbf{i})$ and $\Phi(\mathbf{j})$ were given by Eq. (18).

This completes the proof. □

Remark 2 If we use the matrix parameters $A_1 = A_2 = [0,\ 1;\ -1,\ 0]$, then Eq. (21) will lead to

$$\begin{aligned}
&\mathscr{F}\{f \vartriangle g\}(\omega)\\
&= e^{\mathbf{i}\omega_1}\{[\mathscr{F}\{f_0\}(-\omega)\mathscr{F}\{g_0\}(\omega) + \mathscr{F}\{f_1\}(-\omega)\mathscr{F}\{g_1\}(\omega)\\
&+ \mathscr{F}\{f_2\}(-\omega)\mathscr{F}\{g_2\}(\omega) + \mathscr{F}\{f_3\}(-\omega)\mathscr{F}\{g_3\}(\omega)]\\
&+ \mathbf{i}[\mathscr{F}\{f_0\}(-\omega)\mathscr{F}\{g_1\}(\omega) - \mathscr{F}\{f_1\}(-\omega)\mathscr{F}\{g_0\}(\omega)\\
&- \mathscr{F}\{f_2\}(-\omega)\mathscr{F}\{g_3\}(\omega) + \mathscr{F}\{f_3\}(-\omega)\mathscr{F}\{g_2\}(\omega)]\\
&+ \mathbf{j}[\mathscr{F}\{f_0\}(-\omega)\mathscr{F}\{g_2\}(\omega) - \mathscr{F}\{f_2\}(-\omega)\mathscr{F}\{g_0\}(\omega)\\
&- \mathscr{F}\{f_3\}(-\omega)\mathscr{F}\{g_1\}(\omega) + \mathscr{F}\{f_1\}(-\omega)\mathscr{F}\{g_3\}(\omega)]\\
&+ \mathbf{k}[\mathscr{F}\{f_0\}(-\omega)\mathscr{F}\{g_3\}(\omega) - \mathscr{F}\{f_3\}(-\omega)\mathscr{F}\{g_0\}(\omega)\\
&- \mathscr{F}\{f_1\}(-\omega)\mathscr{F}\{g_2\}(\omega) + \mathscr{F}\{f_2\}(-\omega)\mathscr{F}\{g_1\}(\omega)]\}e^{\mathbf{j}\omega_2}.
\end{aligned} \tag{27}$$

3.3 Product theorem for QLCT

Theorem 3 *Let $f, g \in L^1(\mathbb{R}^2; \mathbb{H})$, and $h(\boldsymbol{x}) = f(\boldsymbol{x})g(\boldsymbol{x})$, the QLCT of the product $h(\boldsymbol{x})$ is given by*

$$\begin{aligned}
&\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{h\}(\omega)\\
&= \frac{1}{2\pi b_1 b_2} e^{i\frac{d_1}{2b_1}\omega_1^2} \cdot \left(\widetilde{\mathscr{F}}\{g\}(\omega) * \mathscr{F}\{f\}\left(\frac{\omega}{\boldsymbol{b}}\right)\right) \cdot e^{j\frac{d_2}{2b_2}\omega_2^2}.
\end{aligned} \tag{28}$$

Proof Based on the definition of QFT and QLCT [30], we have

$$\begin{aligned}
&\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{h\}(\omega)\\
&= \int_{\mathbb{R}^2} K^{\mathbf{i}}_{A_1}(x_1,\omega_1) f(\mathbf{x}) g(\mathbf{x}) K^{\mathbf{j}}_{A_2}(x_2,\omega_2) d\mathbf{x}\\
&= \int_{\mathbb{R}^2} K^{\mathbf{i}}_{A_1}(x_1,\omega_1) f(\mathbf{x}) \cdot \int_{\mathbb{R}^2} K^{-\mathbf{i}}_{A_1}(x_1,v_1) \mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{g\}(\mathbf{v})\\
&\cdot K^{-\mathbf{j}}_{A_2}(x_2,v_2) d\mathbf{v} \cdot K^{\mathbf{j}}_{A_2}(x_2,\omega_2) d\mathbf{x}\\
&= \int_{\mathbb{R}^2}\int_{\mathbb{R}^2} \frac{1}{2\pi b_1} e^{\mathbf{i}\frac{d_1}{2b_1}(\omega_1^2 - v_1^2) - \mathbf{i}\frac{x_1}{b_1}(\omega_1 - v_1)} f(\mathbf{x}) \mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{g\}(\mathbf{v})\\
&\cdot \frac{1}{2\pi b_2} e^{\mathbf{j}\frac{d_2}{2b_2}(\omega_2^2 - v_2^2) - \mathbf{j}\frac{x_2}{b_2}(\omega_2 - v_2)} d\mathbf{x}d\mathbf{v}\\
&= \int_{\mathbb{R}^2} \frac{1}{4\pi^2 b_1 b_2} e^{\mathbf{i}\frac{d_1}{2b_1}(\omega_1^2 - v_1^2)} \int_{\mathbb{R}^2} e^{-\mathbf{i}\frac{x_1}{b_1}(\omega_1 - v_1)} f(\mathbf{x})\\
&\cdot e^{-\mathbf{j}\frac{x_2}{b_2}(\omega_2 - v_2)} d\mathbf{x} \cdot \mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{g\}(\mathbf{v}) e^{\mathbf{j}\frac{d_2}{2b_2}(\omega_2^2 - v_2^2)} d\mathbf{v}\\
&= \int_{\mathbb{R}^2} \frac{1}{2\pi b_1 b_2} e^{\mathbf{i}\frac{d_1}{2b_1}\omega_1^2} \cdot e^{-\mathbf{i}\frac{d_1}{2b_1}v_1^2} \mathscr{F}\{f\}\left(\frac{\omega - \mathbf{v}}{\mathbf{b}}\right)\\
&\cdot \mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{g\}(\mathbf{v}) e^{\mathbf{j}\frac{d_2}{2b_2}\omega_2^2} \cdot e^{-\mathbf{j}\frac{d_2}{2b_2}v_2^2} d\mathbf{v}\\
&= \frac{1}{2\pi b_1 b_2} e^{\mathbf{i}\frac{d_1}{2b_1}\omega_1^2} \cdot \int_{\mathbb{R}^2} e^{-\mathbf{i}\frac{d_1}{2b_1}v_1^2} \mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{g\}(\mathbf{v}) e^{-\mathbf{j}\frac{d_2}{2b_2}v_2^2}\\
&\cdot \mathscr{F}\{f\}\left(\frac{\omega - \mathbf{v}}{\mathbf{b}}\right) d\mathbf{v} \cdot e^{\mathbf{j}\frac{d_2}{2b_2}\omega_2^2}.
\end{aligned} \tag{29}$$

Based on the fundamental relationship between the QLCT and QFT [30], we have

$$\begin{aligned}
&\mathscr{L}^{\mathbb{H}}_{A_1,A_2}\{h\}(\omega) \\
&= \frac{1}{2\pi b_1 b_2} e^{\mathbf{i}\frac{d_1}{2b_1}\omega_1^2} \\
&\quad \cdot \int_{\mathbb{R}^2} \widetilde{\mathscr{F}}\{g\}(\omega)\cdot \mathscr{F}\{f\}\left(\frac{\omega-\mathbf{v}}{\mathbf{b}}\right)\mathrm{d}\mathbf{v}\cdot e^{\mathbf{j}\frac{d_2}{2b_2}\omega_2^2} \\
&= \frac{1}{2\pi b_1 b_2} e^{\mathbf{i}\frac{d_1}{2b_1}\omega_1^2}\cdot\left(\widetilde{\mathscr{F}}\{g\}(\omega) * \mathscr{F}\{f\}\left(\frac{\omega}{\mathbf{b}}\right)\right)\cdot e^{\mathbf{j}\frac{d_2}{2b_2}\omega_2^2}.
\end{aligned} \tag{30}$$

This completes the proof. □

Thus, the product theorem in QLCT domain states that the QLCT of the product of f and g is obtained by conventional convolution between the QFT of f and g.

4 Conclusions

In this paper, motivated by the one-dimensional convolution operator for LCT (see [11]) and based on the quaternion theory, a novel two-dimensional convolution operator for QLCT is proposed in the first. It is shown that our new canonical convolution operator is much more flexible and useful in certain cases. Then, a correlation operator for QLCT which is related to the convolution operator is proposed, and the corresponding correlation theorem is investigated. Finally, based on the definition of the QLCT, the product theorem associated with QLCT is obtained.

Acknowledgements This work is supported by the National Natural Science Foundation of China (No. 61671063) and the Foundation for Innovative Research Groups of the National Natural Science Foundation of China (No. 61421001).

References

1. Koc, A., Ozaktas, H.M., Candan, C., Alper Kutay, M.: Digital computation of linear canonical transforms. IEEE Trans. Signal Process. **56**(6), 2383–2394 (2008)
2. Wei, D.Y., Ran, Q.W., Li, Y.M.: A convolution and correlation theorem for the linear canonical transform and its application. Circuits Syst. Signal Process. **31**(1), 301–312 (2012)
3. Zhao, J., Tao, R., Wang, Y.: Multi-channel filter banks associated with linear canonical transform. Signal Process. **93**(4), 695–705 (2013)
4. Li, Y.C., Zhang, F., Li, Y.B., Tao, R.: Application of linear canonical transform correlation for detection of linear frequency modulated signals. IET Signal Process. **10**(4), 351–358 (2016)
5. Guo, Y., Li, B.Z.: Blind image watermarking method based on linear canonical wavelet transform and QR decomposition. IET Signal Process. **10**(10), 773–786 (2016)
6. Xu, S.Q., Li, F., Yi, C., He, Y.G.: Analysis of A-stationary random signals in the linear canonical transform domain. Signal Process. **146**, 126–132 (2018)
7. Zhao, J., Tao, R., Li, Y.L., Wang, Y.: Uncertainty principles for linear canonical transform. IEEE Trans. Signal Process. **57**(7), 2856–2858 (2009)
8. Wei, D.Y., Li, Y.M.: Sampling and series expansion for linear canonical transform. Signal Image Video Process. **8**(6), 1095–1101 (2014)
9. Huo, H.Y., Sun, W.C.: Sampling theorems and error estimates for random signals in the linear canonical transform domain. Signal Process. **111**, 31–38 (2015)
10. Feng, Q., Li, B.Z.: Convolution and correlation theorems for the two-dimensional linear canonical transform and its applications. IET Signal Process. **10**(2), 125–132 (2016)
11. Huo, H.Y.: A new convolution theorem associated with the linear canonical transform. Signal Image Video Process. **13**, 127–133 (2018)
12. Urynbassarova, D., Li, B.Z., Tao, R.: Convolution and correlation theorems for Wigner–Ville distribution associated with the offset linear canonical transform. Opt. Int. J. Light Electron Opt. **157**, 455–466 (2018)
13. Wei, D.Y.: New product and correlation theorems for the offset linear canonical transform and its applications. Opt. Int. J. Light Electron Opt. **164**, 243–253 (2018)
14. Ell, T.A., Sangwine, S.J.: Hypercomplex Fourier transforms of color images. IEEE Trans. Image Process. **16**(1), 22–35 (2007)
15. Xu, G.L., Tong, W.X., Gang, X.X.: Fractional quaternion Fourier transform, convolution and correlation. Signal Process. **88**(10), 2511–2517 (2008)
16. Kou, K.I., Ou, J.Y., Morais, J.: On uncertainty principle for quaternionic linear canonical transform. In: Abstract and Applied Analysis, vol. 2013, Article ID 725952 (2013)
17. Kou, K.I., Morais, J.: Asymptotic behaviour of the quaternion linear canonical transform and the Bochner–Minlos theorem. Appl. Math. Comput. **247**, 675–688 (2014)
18. Akila, L., Roopkumar, R.: Multidimensional quaternionic Gabor transforms. Appl. Math. Comput. **26**(3), 985–1011 (2016)
19. Fletcher, P., Sangwine, S.J.: The development of the quaternion wavelet transform. Signal Process. **136**, 2–15 (2017)
20. Kou, K.I., Ou, J.Y., Morais, J.: Uncertainty principles associated with quaternionic linear canonical transforms. Math. Methods Appl. Sci. **39**(10), 2722–2736 (2016)
21. Zhang, Y.N., Li, B.Z.: Novel uncertainty principles for two-sided quaternion linear canonical transform. Adv. Appl. Clifford Algebras **28**(1), 15 (2018)
22. Bahri, M., Ashino, R.: Two-dimensional quaternion linear canonical transform: properties, convolution, correlation, and uncertainty principle. J. Math. 2019, Article ID 1062979 (2019)
23. Bahri, M., Ashino, R.: A convolution theorem related to quaternion linear canonical transform. In: Abstract and Applied Analysis, vol. 2019, Article ID 3749387 (2019)
24. Pei, S.C., Ding, J.J.: Relations between fractional operations and time-frequency distributions, and their applications. IEEE Trans. Signal Process. **49**(8), 1638–1655 (2001)
25. Wei, D.Y., Ran, Q.W., Li, Y.M., Ma, J., Tan, L.Y.: A convolution and product theorem for the linear canonical transform. IEEE Signal Process. Lett. **16**(10), 853–856 (2009)
26. Singh, A.K., Saxena, R.: On convolution and product theorems for FRFT. Wirel. Pers. Commun. **65**(1), 189–201 (2012)

27. Shi, J., Liu, X.P., Zhang, N.T.: Generalized convolution and product theorems associated with linear canonical transform. Signal Image Video Process. **8**(5), 967–974 (2014)
28. Feng, Q., Wang, R.B.: Fractional convolution, correlation theorem and its application in filter design. Signal Image Video Process. **14**, 351–358 (2019)
29. Hamilton, W.R.: Elements of Quaternions. Longmans green, London (1866)
30. Bahri, M., Ashino, R.: A simplified proof of uncertainty principle for quaternion linear canonical transform. In: Abstract and Applied Analysis, vol. 2016(6), pp. 1–11 (2016)